EGX Stock Market - Data Acquisition

EGX Stock Market — Summary Report

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Dataset: Egyptian Exchange (EGX) — 32 Shariah-Compliant Companies
Period: June 2025 – December 2025
Records: ~4,640

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1. Overview

This project presents a complete data science pipeline for analyzing Shariah-compliant companies listed on the Egyptian Exchange (EGX).

It includes data collection, preprocessing, feature engineering, and financial analysis to understand market behavior, risk, and performance across companies and sectors.

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2. Data Collection

Sources

- **yfinance API** → historical stock data (OHLCV)
- **BeautifulSoup** → static scraping
- **Selenium** → dynamic scraping (African Markets)

Challenges

- Google Finance (SerpApi) does not support EGX
- Manual discovery of ticker symbols (`.CA`)
- Inconsistent company naming across datasets

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3. Data Preprocessing

Key steps:

- Removed irrelevant columns (zero values)
- Cleaned column names
- Converted **Date** to datetime format

- Checked and handled missing values
- Standardized company names using mapping
- Converted text data to numeric values

Result: Clean and consistent dataset ready for analysis

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4. Feature Engineering

Main calculated features:

- **Returns (%)** → performance measurement
- **Volatility** → risk estimation
- **Price Range** → daily movement
- **Portfolio Value** → cumulative growth simulation

These features enabled comparison across companies and sectors.

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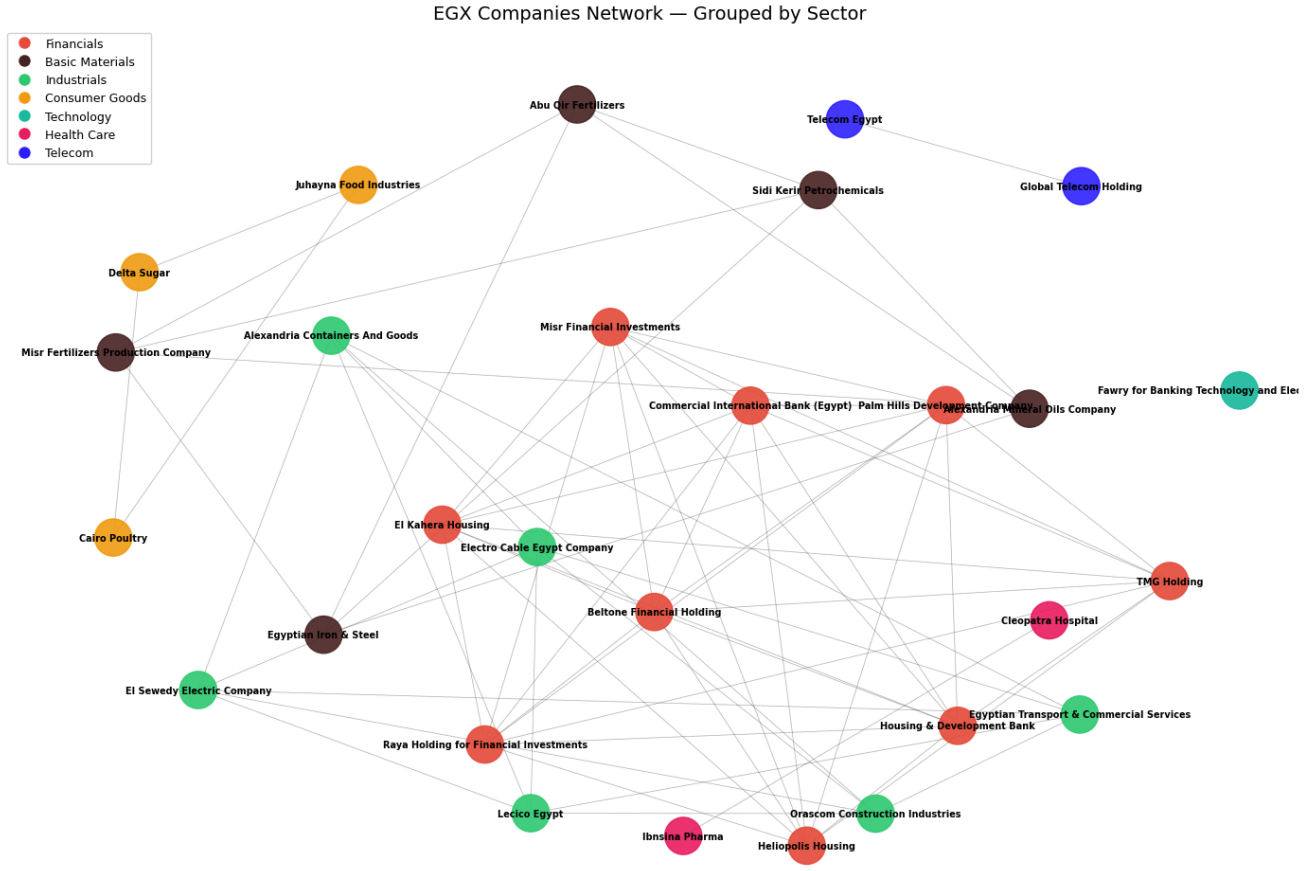
5. Network Analysis

- Companies represented as **nodes**
- Connections based on **sector grouping**

Insight

- Strong clustering within sectors
- No connections between different sectors

Visualization



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6. Sector Analysis & Portfolio Simulation

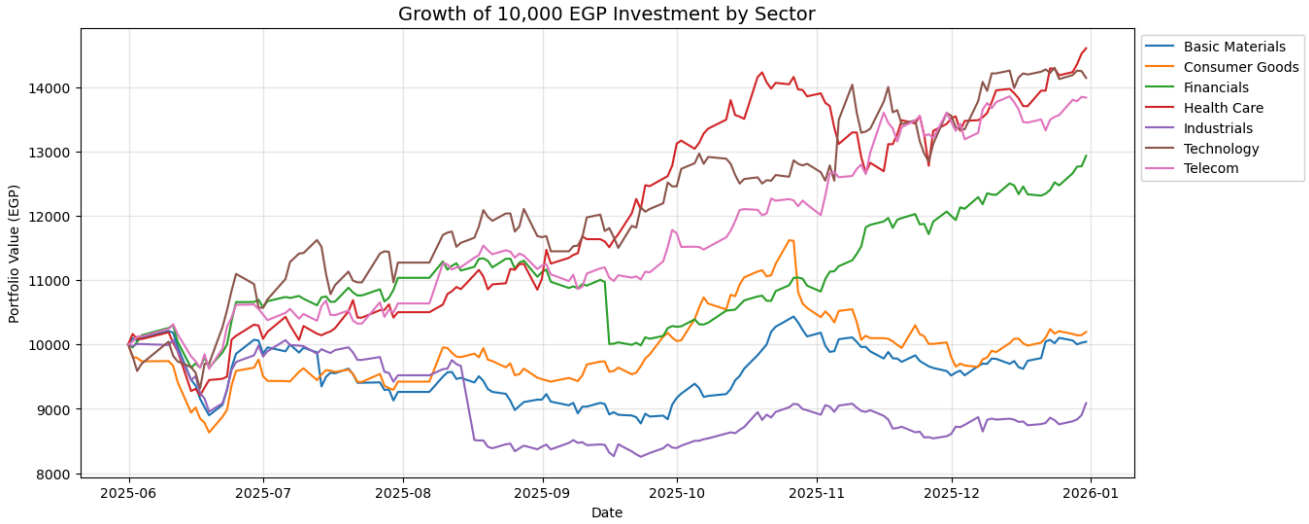
Method

- Initial investment: **10,000 EGP**
- Returns averaged per sector
- Compounded over time

Result

- **Top performing sectors:**
  - Healthcare
  - Technology
  - Telecom
- **Lower performing sectors:**
  - Basic Materials
  - Consumer Goods

Visualization

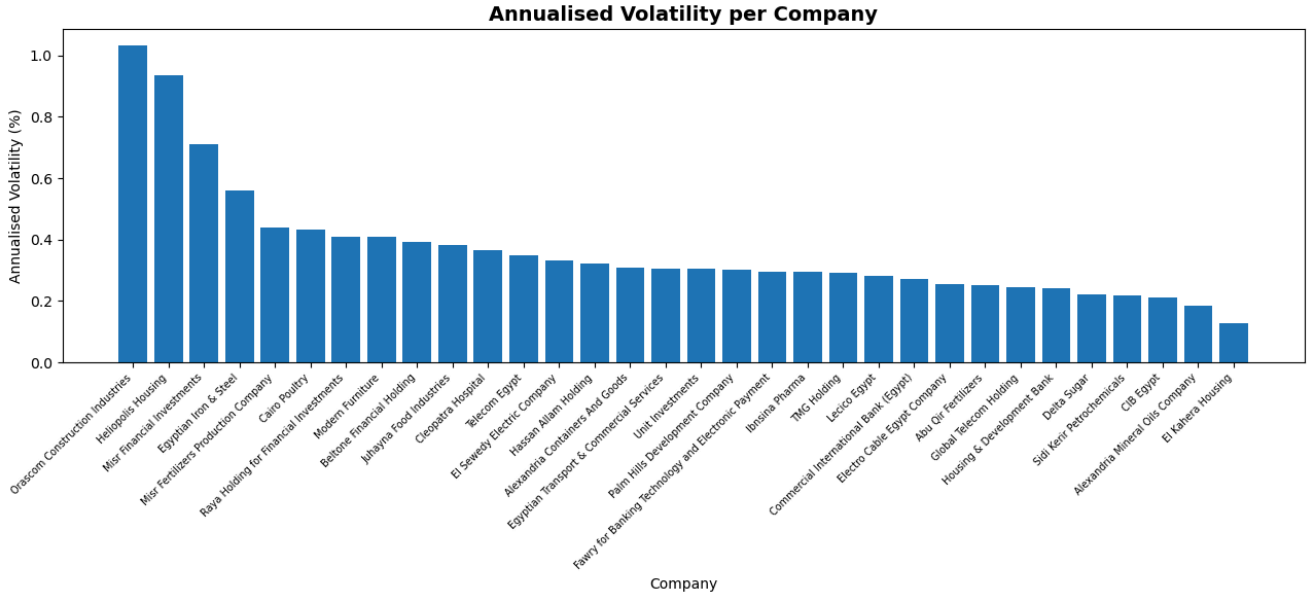


7. Risk Analysis (Volatility)

Key Findings

- **High Risk Companies:**
  - Orascom Construction
  - Misr Fertilizers
  - Heliopolis Housing
- **Low Risk Companies:**
  - El Kahera Housing
  - Sidi Kerir Petrochemicals
  - Housing & Development Bank

Visualization



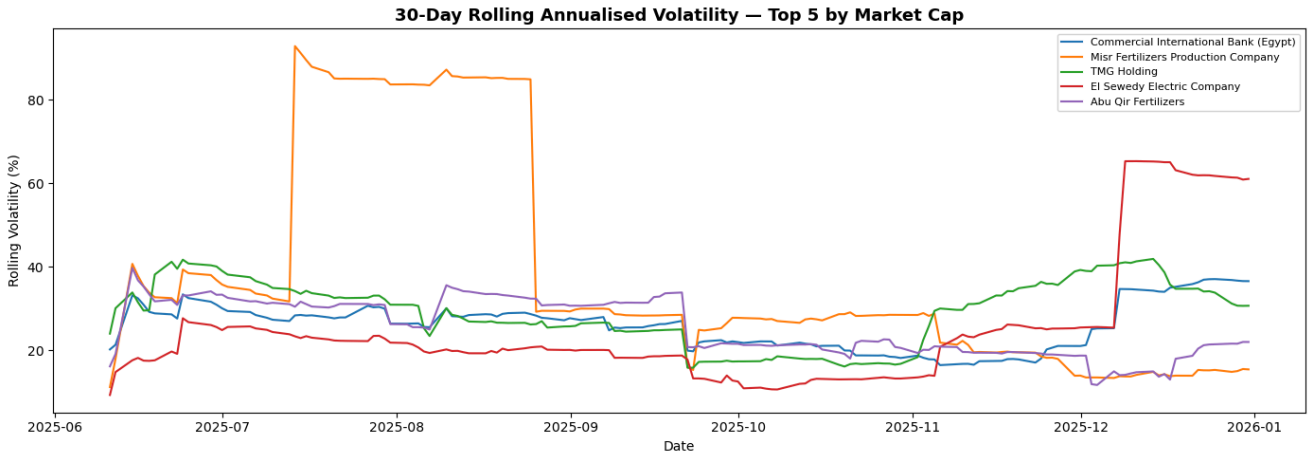
8. Rolling Volatility (Top Companies)

- 30-day rolling volatility used
- Shows how risk changes over time

Insight

- Misr Fertilizers → highest fluctuations
- CIB & El Sewedy → more stable

Visualization



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9. Key Insights

- Market performance varies across sectors
- Higher returns often come with higher risk
- Large companies are not always stable
- Sector choice strongly affects portfolio results

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10. Conclusion

This project demonstrates a full data acquisition workflow applied to financial markets, starting from data collection challenges to building meaningful financial insights.